Suicide-Risk Detection Using NLP-Based Text and Speech Analysis

A Project Report

Submitted in partial fulfilment of requirements for the award of the degree of
BACHELOR OF TECHNOLOGY
IN
INFORMATION TECHNOLOGY

Submitted by
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DEPARTMENT OF INFORMATION TECHNOLOGY
LAKIREDDY BALI REDDY COLLEGE OF ENGINEERING
(AUTONOMOUS)
An ISO 21001:2018, 14001:2015, 50001:2018 Certified Institution
Approved by AICTE, New Delhi and Affiliated to JNTUK, Kakinada
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2026-2027

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DEPARTMENT OF INFORMATION TECHNOLOGY

CERTIFICATE

This is to certify that S. Thanmayi (23761A1251), Ch. Venu Madhav (24765A1201) and M. Samuel Babu (23761A1235), students of IV B. Tech (Information Technology), have successfully completed the project titled "Suicide-Risk Detection Using NLP-Based Text and Speech Analysis" at Lakireddy Bali Reddy College of Engineering during the Academic Year 2026-2027. This project report is submitted as partial fulfilment for the award of the degree of Bachelor of Technology (Information Technology).

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We are also thankful to the project coordinator, Mr. M. Vijay Kumar, Sr Assistant Professor, Department of IT, for coordinating the project reviews and providing systematic guidance.

Finally, we express our gratitude to our parents and friends for their constant encouragement and support.

CANDIDATE'S DECLARATION

We hereby declare that the project report entitled "Suicide-Risk Detection Using NLP-Based Text and Speech Analysis" is a record of our work carried out during the academic year 2026-2027 under the guidance of Dr. G. Rajendra, Associate Professor, Department of Information Technology, and submitted in partial fulfilment of the requirements for the award of the degree of Bachelor of Technology in Information Technology.

The matter embodied in this project is genuine work carried out by us and has not been submitted earlier to this university or any other university for the award of any degree, diploma or prize.

PROJECT ASSOCIATES
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ACRONYMS AND ABBREVIATIONS

AI	Artificial Intelligence
NLP	Natural Language Processing
SDCNL	Suicide and Depression Classification with Noisy Labels
EATD	Emotional Audio-Textual Depression Corpus
TF-IDF	Term Frequency-Inverse Document Frequency
SVM	Support Vector Machine
BERT	Bidirectional Encoder Representations from Transformers
BiLSTM	Bidirectional Long Short-Term Memory
LR	Logistic Regression
MFCC	Mel-Frequency Cepstral Coefficients
RMS	Root Mean Square Energy
ZCR	Zero-Crossing Rate
ROC-AUC	Receiver Operating Characteristic - Area Under Curve
CV	Cross-Validation
UI	User Interface

ABSTRACT

Suicide-risk identification is a challenging research problem because warning signals can appear in written language, speech characteristics, or both. This project presents a multimodal research screening framework that combines Natural Language Processing based text analysis with acoustic speech analysis. The text branch uses the SDCNL dataset and evaluates a BiLSTM, frozen BERT mean-pooled embeddings with Logistic Regression, and TF-IDF bigram features with a calibrated Linear SVM. After leakage-free preprocessing, the final SDCNL training set contains 1,513 records and the held-out test set contains 379 records, with zero normalized train-test text overlap. The calibrated Linear SVM is the best verified text model with 72.56% accuracy, 71.92% precision, 75.65% recall, 73.74% F1-score and 78.22% ROC-AUC. The speech branch uses the EATD-Corpus as a complementary depression/distress source. Speech is processed at mono 16 kHz with peak normalization, and acoustic features including MFCC, pitch, RMS energy, zero-crossing rate, spectral centroid and duration are extracted. A participant-grouped Logistic Regression speech model achieved 63.29% participant-level validation accuracy, 63.44% balanced accuracy and 59.22% ROC-AUC. Since SDCNL and EATD contain different participants and different targets, they are not merged row-by-row. The final application keeps P_text and P_speech separate and applies a transparent prototype late-fusion decision layer for Low, Moderate and High research-screening levels. The Streamlit implementation supports text-only, speech-only and combined input modes, and the final automated verification suite passed 37 out of 37 checks. The system is intended for academic research screening and not for clinical diagnosis.

CHAPTER - 1

1.1. INTRODUCTION:

Suicide and self-harm related risk is a major mental-health concern, and early identification of warning signals can support timely human intervention. Conventional assessment depends on professional interviews and validated clinical instruments. These remain essential for diagnosis, but they are difficult to scale continuously across large online populations.

Digital platforms have created new sources of behavioural and linguistic information. Natural Language Processing can model text patterns, while speech processing can capture complementary acoustic characteristics such as pitch variation, energy and spectral patterns.

This project studies a modular multimodal architecture with independent SDCNL text and EATD speech branches. Because the datasets contain different participants and different targets, only late decision-level fusion is used.

1.2. EXISTING SYSTEM:

• Traditional clinical screening requires expert time and availability.

• Automated text systems can be affected by noisy labels, leakage and calibration problems.

• Public speech datasets with direct suicide-risk labels are scarce.

• Unmatched datasets cannot be safely concatenated as if they were paired multimodal records.

1.3. PROPOSED SYSTEM:

• Leakage-free SDCNL text pipeline.

• BiLSTM, frozen BERT + LR, and calibrated TF-IDF SVM comparison.

• EATD acoustic speech model with participant-grouped validation.

• Late fusion of P_text and P_speech only.

• Streamlit UI with Text, Speech and Both modes.

• Automated 37-check final verification.

CHAPTER - 2

2. LITERATURE SURVEY

Haque, Reddi and Giallanza (ICANN 2021): proposed SDCNL for suicide-versus-depression text classification and addressed noisy web labels using unsupervised label correction. Their work is directly relevant to the text dataset used in this project and motivates careful treatment of noisy labels and evaluation leakage.

Shen, Yang and Lin (ICASSP 2022): introduced the EATD-Corpus and a GRU/BiLSTM-based depression detection framework using emotional audio and textual responses. EATD contains audio and transcripts from 162 volunteers and provides the speech/distress source used in this project.

Pokrywka, Kaczmarek and Gorzelańczyk (IEEE BigData 2024): evaluated modern transformer configurations for suicide-risk classification on social-media text and demonstrated the strong potential of contextual language models in risk classification.

Jin, Liu and Dai (2025): studied multimodal speech and text models for adolescent suicide-risk detection using paired recordings and transcripts. Their findings support multimodal analysis, but their paired-data setting differs from this project because SDCNL and EATD are unmatched.

Devlin et al. (2019): introduced BERT, a bidirectional transformer architecture that enables contextual representation of language. In this project BERT is used as a frozen mean-pooled embedding baseline rather than as a fully fine-tuned final model.

Classical calibrated machine learning: remains competitive on limited text datasets. The project therefore compares deep-learning representations against a calibrated linear SVM with TF-IDF features instead of assuming a transformer will automatically provide the best held-out performance.

Research Gap

Existing research shows the value of text, speech and multimodal analysis, but the public datasets available to this project are not participant-matched. The project therefore emphasizes leakage-free branch evaluation and transparent decision-level fusion rather than unsupported feature-level merging.

CHAPTER - 3

3. SYSTEM SPECIFICATIONS

3.1 SOFTWARE REQUIREMENTS:

Python 3.11: Primary programming environment used for preprocessing, model training, evaluation, inference and testing.
NumPy: Numerical operations and feature arrays.
Pandas: CSV loading, cleaning, participant-level feature tables and evaluation data.
Scikit-learn: TF-IDF, LinearSVC, probability calibration, Logistic Regression, feature selection, cross-validation and metrics.
PyTorch: BiLSTM text baseline implementation.
Hugging Face Transformers: Frozen BERT embedding baseline.
Librosa: Audio loading and acoustic feature extraction.
SoundFile: Audio support.
Joblib: Saving and loading vectorizers and trained models.
Streamlit: Local browser-based user interface.
Matplotlib: Charts and evaluation figures.

3.2 HARDWARE REQUIREMENTS:

• Windows 10/11 64-bit

• Intel Core i5/i7 or equivalent

• 8 GB RAM minimum; 16 GB preferred

• 10-20 GB free storage

• GPU optional for final inference

3.3 FUNCTIONAL REQUIREMENTS:

- Text-only, speech-only and combined input
- Saved-model inference
- Audio feature extraction
- Prototype fusion
- Screening result display
- Reusable artifacts without retraining

3.4 NON-FUNCTIONAL REQUIREMENTS:

- Usability
- Reliability
- Maintainability
- Transparency
- Reproducibility
- Safe interpretation

CHAPTER - 4

4. METHODOLOGY

4.1 Overview of the Project

The system contains independent text and speech branches followed by a transparent prototype decision layer.

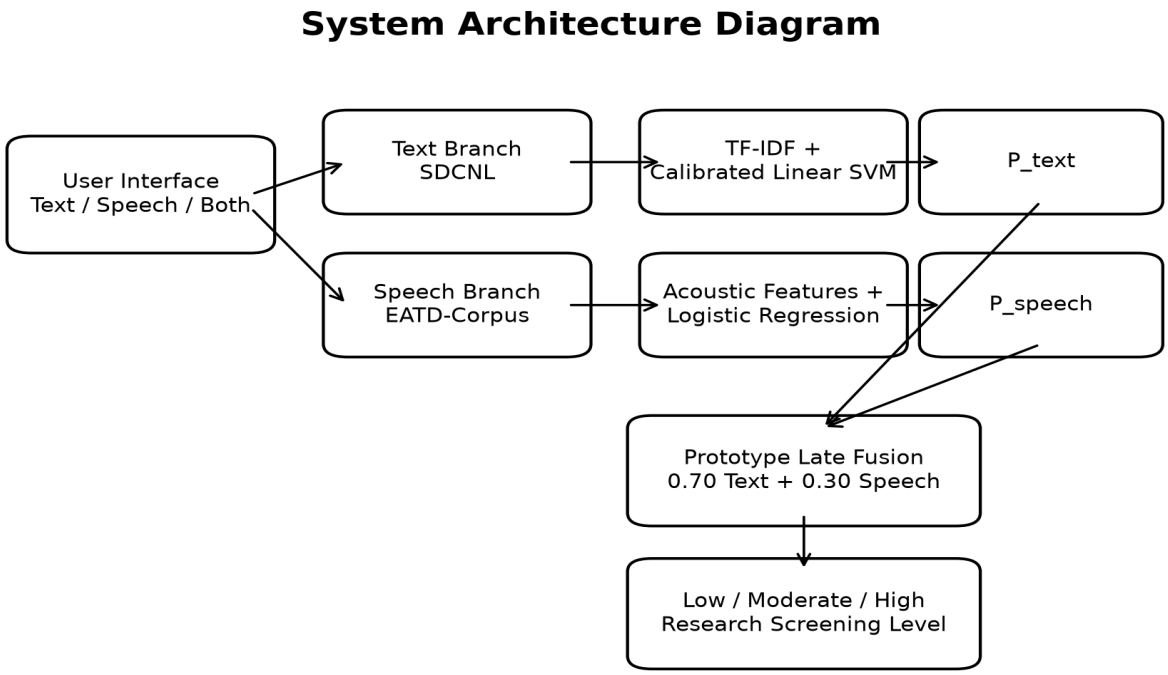


Figure 4.1: System Architecture Diagram

4.2 Text Data Pre-processing

• clean model_text

• remove empty inputs

• remove exact duplicates

• preserve labels

• check normalized overlap

• keep held-out test isolated

Final text split: 1,513 training rows and 379 held-out test rows with zero normalized overlap.

4.3 Text Model Development

• BiLSTM baseline

• TF-IDF + calibrated Linear SVM

• Frozen BERT embeddings + Logistic Regression

• Same held-out test set for all comparisons

4.4 Speech Pre-processing and Feature Extraction

Audio is loaded at 16 kHz mono, peak normalized, and represented using MFCC, pitch, RMS, ZCR, spectral centroid and duration features.

4.5 Speech Model Selection

Participant-grouped cross-validation prevents participant leakage across folds. Official EATD validation is used only after selection.

4.6 Prototype Late Fusion

Combined = 0.70 × P_text + 0.30 × P_speech. Low < 0.40, Moderate < 0.70, High ≥ 0.70. These are prototype research settings, not clinical thresholds.

CHAPTER - 5

5. DESIGN VIEW

5.1 System Flow Chart

The flow chart describes how the user selects an input mode, provides the required data, triggers branch inference, and receives the final research-screening result.

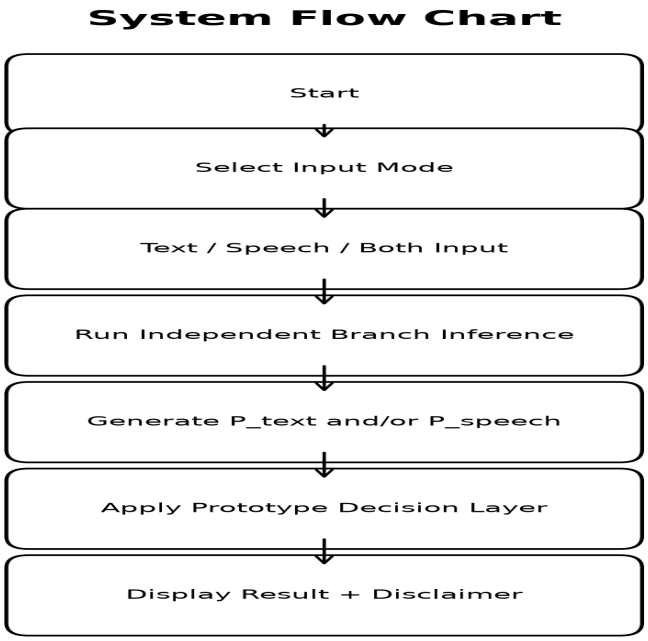


Figure 5.1: System Flow Chart

5.2 Logical Diagram

The logical view separates the input, processing, model and decision layers so that text and speech processing remain independent until the final decision stage.

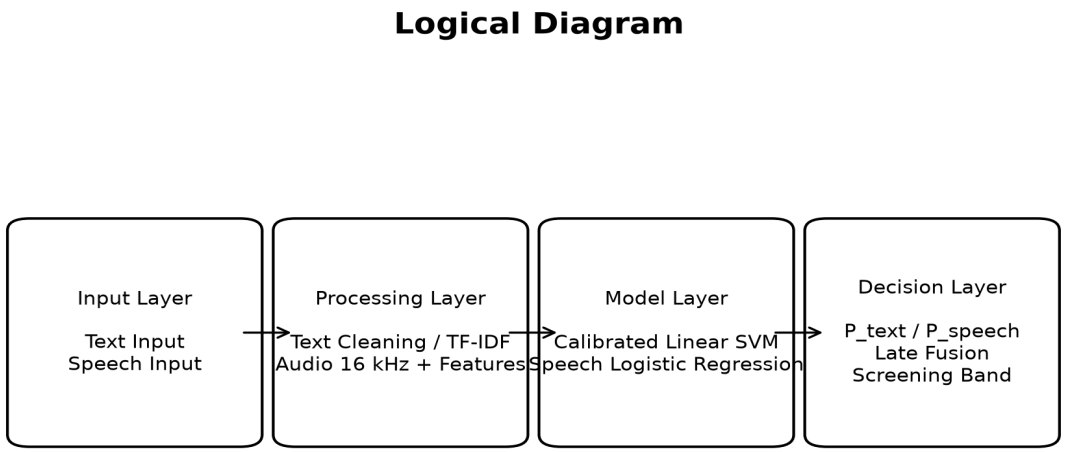


Figure 5.2: Logical Diagram

5.3 Behavioural / Activity Diagram

The activity diagram represents the application workflow from input selection through validation, model execution, score combination and output display.

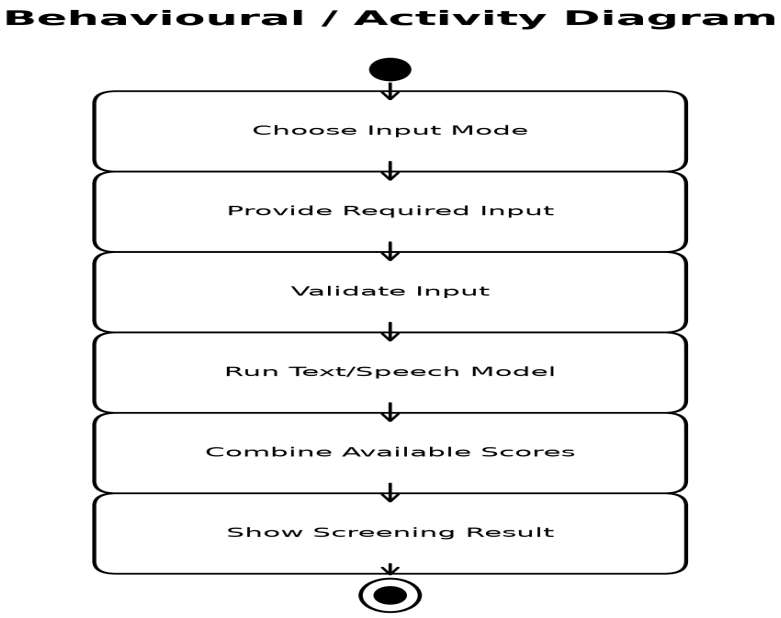


Figure 5.3: Activity Diagram

5.4 Conceptual ER Diagram

The ER view is conceptual because the current prototype uses files and saved model artifacts rather than a relational database. It shows the logical relationship among input, branch results and screening result.

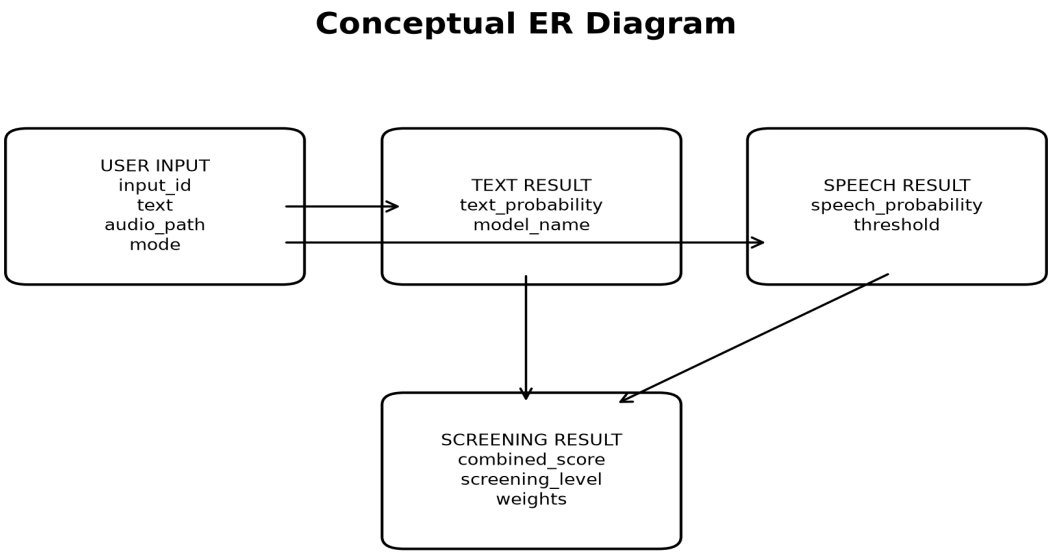


Figure 5.4: Conceptual ER Diagram

5.5 Data Flow Diagram

The DFD shows user input entering the text and speech analysis processes, branch scores flowing into the decision process and the final result returning to the user.

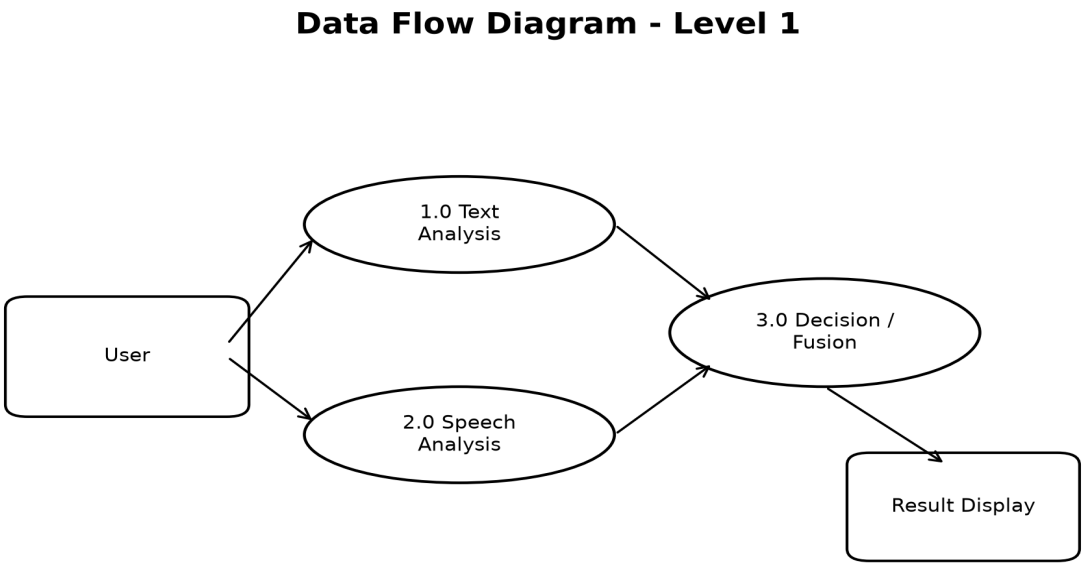


Figure 5.5: DFD Level 1

CHAPTER - 6

6. IMPLEMENTATION

The implementation is organized into dataset preprocessing, independent model training, saved model artifacts, a unified inference pipeline, a Streamlit user interface and automated verification. The following pages show the principal source-code components from the verified project. The complete executable source is stored in the project folder.

6.1 Text SVM Training

```
import json
from pathlib import Path
import pandas as pd
from sklearn.feature_extraction.text import TfidfVectorizer
from sklearn.svm import LinearSVC
from sklearn.metrics import accuracy_score, precision_recall_fscore_support, confusion_matrix, roc_auc_score, classification_report
from sklearn.calibration import CalibratedClassifierCV
import joblib

BASE=Path(r"C:\Users\RAMESH\Desktop\Projects\Thanmayi 8000")
DATA=BASE/"Datasets"
MODEL_DIR=BASE/"models"/"text_model"
REPORT_DIR=BASE/"reports"/"text_model"
train=pd.read_csv(DATA/"training-set-clean.csv").dropna(subset=["model_text","is_suicide"])
test=pd.read_csv(DATA/"testing-set-clean-leakage-free.csv").dropna(subset=["model_text","is_suicide"])

vec=TfidfVectorizer(ngram_range=(1,2),min_df=2,max_df=0.98,max_features=50000,sublinear_tf=True)
Xtr=vec.fit_transform(train["model_text"].astype(str))
Xte=vec.transform(test["model_text"].astype(str))
base=LinearSVC(C=1.0,class_weight=None,random_state=42)
clf=CalibratedClassifierCV(base,cv=5,method="sigmoid")
clf.fit(Xtr,train["is_suicide"].astype(int))
pred=clf.predict(Xte)
probs=clf.predict_proba(Xte)[:,1]
y=test["is_suicide"].astype(int).values
acc=accuracy_score(y,pred)
p,r,f,_=precision_recall_fscore_support(y,pred,average="binary",zero_division=0)
auc=roc_auc_score(y,probs)
cm=confusion_matrix(y,pred)
metrics={"model":"TF-IDF bigram + calibrated LinearSVC","accuracy":float(acc),"precision":float(p),"recall":float(r),"f1_score":float(f),"roc_auc":float(auc),"confusion_matri
joblib.dump(vec,MODEL_DIR/"tfidf_vectorizer.joblib")
joblib.dump(clf,MODEL_DIR/"text_svm_calibrated.joblib")
with open(REPORT_DIR/"text_svm_metrics.json","w",encoding="utf-8") as fp: json.dump(metrics,fp,indent=2)
with open(REPORT_DIR/"text_svm_classification_report.txt","w",encoding="utf-8") as fp: fp.write(classification_report(y,pred,digits=4,zero_division=0))
print(json.dumps(metrics,indent=2))
```

6.2 BiLSTM Text Baseline

```
import os, json, random, re
from pathlib import Path
from collections import Counter

import numpy as np
import pandas as pd
import torch
import torch.nn as nn
from torch.utils.data import TensorDataset, DataLoader
from sklearn.model_selection import train_test_split
from sklearn.metrics import accuracy_score, precision_recall_fscore_support, confusion_matrix, roc_auc_score, classification_report
import matplotlib.pyplot as plt

SEED = 42
random.seed(SEED)
np.random.seed(SEED)
torch.manual_seed(SEED)

BASE = Path(r"C:\Users\RAMESH\Desktop\Projects\Thanmayi 8000")
DATA = BASE / "Datasets"
MODEL_DIR = BASE / "models" / "text_model"
REPORT_DIR = BASE / "reports" / "text_model"
MODEL_DIR.mkdir(parents=True, exist_ok=True)
REPORT_DIR.mkdir(parents=True, exist_ok=True)

TRAIN_CSV = DATA / "training-set-clean.csv"
TEST_CSV = DATA / "testing-set-clean-leakage-free.csv"

VOCAB_SIZE = 20000
EMBED_DIM = 128
LSTM_UNITS = 64
BATCH_SIZE = 32
EPOCHS = 20
VAL_SIZE = 0.15
PATIENCE = 3

DEVICE = torch.device("cuda" if torch.cuda.is_available() else "cpu")

def tokenize(text):
    return str(text).strip().split()

def load_data():
    train_df = pd.read_csv(TRAIN_CSV)
    test_df = pd.read_csv(TEST_CSV)
    required = {"model_text", "is_suicide"}
    if not required.issubset(train_df.columns) or not required.issubset(test_df.columns):
        raise ValueError("Required columns model_text and is_suicide are missing.")
    train_df = train_df.dropna(subset=["model_text", "is_suicide"]).copy()
    test_df = test_df.dropna(subset=["model_text", "is_suicide"]).copy()
    train_df["model_text"] = train_df["model_text"].astype(str)
    test_df["model_text"] = test_df["model_text"].astype(str)
    train_df["is_suicide"] = train_df["is_suicide"].astype(int)
    test_df["is_suicide"] = test_df["is_suicide"].astype(int)
    return train_df, test_df

def build_vocab(texts):
    counter = Counter()
    for t in texts:
        counter.update(tokenize(t))
    vocab = {"<PAD>":0, "<OOV>":1}
    for word, _ in counter.most_common(VOCAB_SIZE - 2):
        vocab[word] = len(vocab)
    return vocab

def encode(text, vocab, max_len):
    ids = [vocab.get(tok, 1) for tok in tokenize(text)]
    ids = ids[:max_len]
    length = max(1, len(ids))
    if len(ids) < max_len:
        ids += [0] * (max_len - len(ids))
    return ids, length

class TextBiLSTM(nn.Module):
    def __init__(self, vocab_size, embed_dim, hidden_dim):
        super().__init__()
        self.embedding = nn.Embedding(vocab_size, embed_dim, padding_idx=0)
        self.lstm = nn.LSTM(embed_dim, hidden_dim, batch_first=True, bidirectional=True)
        self.dropout = nn.Dropout(0.35)
        self.fc1 = nn.Linear(hidden_dim * 2, 64)
        self.relu = nn.ReLU()
        self.fc2 = nn.Linear(64, 1)

    def forward(self, x, lengths):
        emb = self.embedding(x)
        packed = nn.utils.rnn.pack_padded_sequence(
            emb, lengths.cpu(), batch_first=True, enforce_sorted=False
        )
        _, (h, _) = self.lstm(packed)
        hcat = torch.cat((h[-2], h[-1]), dim=1)
        z = self.dropout(self.relu(self.fc1(hcat)))
```

6.3 Frozen BERT Embedding Baseline

```
import json, joblib
from pathlib import Path
import numpy as np
import pandas as pd
import torch
from transformers import AutoTokenizer, AutoModel
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import accuracy_score, precision_recall_fscore_support, confusion_matrix, roc_auc_score, classification_report

BASE=Path(r"C:\Users\RAMESH\Desktop\Projects\Thanmayi 8000")
DATA=BASE/"Datasets"
MODEL_DIR=BASE/"models"/"text_model"
REPORT_DIR=BASE/"reports"/"text_model"
CACHE=Path(r"C:\Users\RAMESH\.cache\huggingface\hub\models--bert-base-uncased\snapshots\86b5e0934494bd15c9632b12f734a8a67f723594")
MODEL_DIR.mkdir(parents=True,exist_ok=True)
REPORT_DIR.mkdir(parents=True,exist_ok=True)

train=pd.read_csv(DATA/"training-set-clean.csv").dropna(subset=["model_text","is_suicide"]).copy()
test=pd.read_csv(DATA/"testing-set-clean-leakage-free.csv").dropna(subset=["model_text","is_suicide"]).copy()
device=torch.device("cpu")
tokenizer=AutoTokenizer.from_pretrained(str(CACHE),local_files_only=True)
encoder=AutoModel.from_pretrained(str(CACHE),local_files_only=True).to(device)
encoder.eval()

def embed_texts(texts,batch_size=16,max_len=128):
    out=[]
    texts=list(map(str,texts))
    for i in range(0,len(texts),batch_size):
        batch=texts[i:i+batch_size]
        enc=tokenizer(batch,padding=True,truncation=True,max_length=max_len,return_tensors="pt")
        enc={k:v.to(device) for k,v in enc.items()}
        with torch.no_grad():
            res=encoder(**enc).last_hidden_state
            mask=enc["attention_mask"].unsqueeze(-1).float()
            pooled=(res*mask).sum(1)/mask.sum(1).clamp(min=1e-9)
        out.append(pooled.cpu().numpy())
        print(f"embedded {min(i+batch_size,len(texts))}/{len(texts)}",flush=True)
    return np.vstack(out)

Xtr=embed_texts(train["model_text"].values)
Xte=embed_texts(test["model_text"].values)

np.save(MODEL_DIR/"bert_train_embeddings.npy",Xtr)
np.save(MODEL_DIR/"bert_test_embeddings.npy",Xte)

clf=LogisticRegression(max_iter=3000,C=1.0,random_state=42)
clf.fit(Xtr,train["is_suicide"].astype(int).values)

y=test["is_suicide"].astype(int).values
pred=clf.predict(Xte)
probs=clf.predict_proba(Xte)[:,1]
acc=accuracy_score(y,pred)
p,r,f,_=precision_recall_fscore_support(y,pred,average="binary",zero_division=0)
auc=roc_auc_score(y,probs)
cm=confusion_matrix(y,pred)

metrics={
 "model":"bert-base-uncased mean-pooled frozen embeddings + LogisticRegression",
 "embedding_model":"bert-base-uncased",
 "max_length":128,
 "train_rows":int(len(train)),
 "test_rows":int(len(test)),
 "accuracy":float(acc),"precision":float(p),"recall":float(r),"f1_score":float(f),"roc_auc":float(auc),
 "confusion_matrix":cm.tolist()
}
joblib.dump(clf,MODEL_DIR/"bert_embedding_logreg.joblib")
with open(MODEL_DIR/"bert_embedding_config.json","w",encoding="utf-8") as fp:
    json.dump({"local_model_path":str(CACHE),"pooling":"attention-mask mean pooling","classifier":"LogisticRegression","test_policy":"held-out leakage-free test only"},fp,ind
with open(REPORT_DIR/"bert_embedding_metrics.json","w",encoding="utf-8") as fp: json.dump(metrics,fp,indent=2)
with open(REPORT_DIR/"bert_embedding_classification_report.txt","w",encoding="utf-8") as fp: fp.write(classification_report(y,pred,digits=4,zero_division=0))
print(json.dumps(metrics,indent=2),flush=True)
```

6.4 Speech Single-Audio Model

```
import json, joblib
from pathlib import Path
import numpy as np
import pandas as pd
from sklearn.pipeline import Pipeline
from sklearn.impute import SimpleImputer
from sklearn.preprocessing import StandardScaler
from sklearn.feature_selection import SelectKBest, f_classif
from sklearn.linear_model import LogisticRegression
from sklearn.svm import SVC
from sklearn.model_selection import StratifiedGroupKFold, cross_val_predict
from sklearn.metrics import accuracy_score, precision_recall_fscore_support, confusion_matrix, roc_auc_score, balanced_accuracy_score, classification_report

BASE=Path(r"C:\Users\RAMESH\Desktop\Projects\Thanmayi 8000")
DATA=BASE/"Datasets"/"Processed_EATD"
MODEL_DIR=BASE/"models"/"speech_model"
REPORT_DIR=BASE/"reports"/"speech_model"
MODEL_DIR.mkdir(parents=True,exist_ok=True)
REPORT_DIR.mkdir(parents=True,exist_ok=True)

train=pd.read_csv(DATA/"EATD-train-features.csv",encoding="utf-8-sig")
val=pd.read_csv(DATA/"EATD-validation-features.csv",encoding="utf-8-sig")
TARGET="depression_label"
GROUP="participant_id"
EXCLUDE={GROUP,"raw_sds_score","standard_sds_score",TARGET,"emotion","transcript","audio_file","audio_source_type",
         "original_duration_sec","original_sample_rate","original_channels","sample_rate_model"}
features=[c for c in train.columns if c not in EXCLUDE and pd.api.types.is_numeric_dtype(train[c]) and train[c].nunique(dropna=True)>1]

X=train[features].copy(); y=train[TARGET].astype(int).values; groups=train[GROUP].astype(str).values
Xv=val[features].copy(); yv=val[TARGET].astype(int).values; gv=val[GROUP].astype(str).values

def binary_metrics(y_true,probs,thr):
    pred=(probs>=thr).astype(int)
    p,r,f,_=precision_recall_fscore_support(y_true,pred,average="binary",zero_division=0)
    return {
      "threshold":float(thr),
      "accuracy":float(accuracy_score(y_true,pred)),
      "balanced_accuracy":float(balanced_accuracy_score(y_true,pred)),
      "precision":float(p),"recall":float(r),"f1_score":float(f),
      "roc_auc":float(roc_auc_score(y_true,probs)),
      "confusion_matrix":confusion_matrix(y_true,pred).tolist()
    }

def participant_aggregate(df, probs):
    tmp=pd.DataFrame({
      "participant_id":df[GROUP].astype(str).values,
      "true_label":df[TARGET].astype(int).values,
      "probability":probs
    })
    return tmp.groupby("participant_id",as_index=False).agg(
      true_label=("true_label","first"), probability=("probability","mean")
    )

cv=StratifiedGroupKFold(n_splits=5,shuffle=True,random_state=42)
specs=[]
for k in [8,12,16,20,25,30,35]:
    if k>len(features): continue
    for C in [0.2,0.5,1.0]:
        specs.append((f"lr_k{k}_c{C}",Pipeline([
          ("imputer",SimpleImputer(strategy="median")),
          ("scaler",StandardScaler()),
          ("select",SelectKBest(f_classif,k=k)),
          ("model",LogisticRegression(max_iter=5000,class_weight="balanced",C=C,random_state=42))
        ])))
    for C in [0.5,1.0,2.0]:
        specs.append((f"rbf_k{k}_c{C}",Pipeline([
          ("imputer",SimpleImputer(strategy="median")),
          ("scaler",StandardScaler()),
          ("select",SelectKBest(f_classif,k=k)),
          ("model",SVC(C=C,gamma="scale",class_weight="balanced",probability=True,random_state=42))
        ])))

candidate_results={}
best=None
for name,pipe in specs:
    oof=cross_val_predict(pipe,X,y,cv=cv,groups=groups,method="predict_proba",n_jobs=1)[:,1]
    agg=participant_aggregate(train,oof)
    yy=agg["true_label"].values; pp=agg["probability"].values
    best_m=None
    for thr in np.linspace(0.15,0.75,121):
        m=binary_metrics(yy,pp,thr)
        score=(m["f1_score"],m["balanced_accuracy"],m["roc_auc"],m["recall"])
        if best_m is None or score>(best_m["f1_score"],best_m["balanced_accuracy"],best_m["roc_auc"],best_m["recall"]):
            best_m=m
    candidate_results[name]=best_m
    rank=(best_m["f1_score"],best_m["balanced_accuracy"],best_m["roc_auc"])
    if best is None or rank>best[0]:
        best=(rank,name,pipe,best_m["threshold"],best_m)

_,best_name,best_pipe,best_thr,best_oof=best
best_pipe.fit(X,y)
val_sample_probs=best_pipe.predict_proba(Xv)[:,1]
val_sample_metrics=binary_metrics(yv,val_sample_probs,best_thr)
val_part=participant_aggregate(val,val_sample_probs)
val_part_metrics=binary_metrics(val_part["true_label"].values,val_part["probability"].values,best_thr)
val_part_default=binary_metrics(val_part["true_label"].values,val_part["probability"].values,0.5)

joblib.dump(best_pipe,MODEL_DIR/"best_speech_single_audio_model.joblib")
selected=[features[i] for i in best_pipe.named_steps["select"].get_support(indices=True)]
config={
 "model_scope":"single audio acoustic depression/distress estimator",
 "best_model":best_name,"threshold":best_thr,
 "all_input_features":features,"selected_features":selected,
```

```
 "train_audio_rows":len(train),"validation_audio_rows":len(val),
 "train_participants":int(train[GROUP].nunique()),"validation_participants":int(val[GROUP].nunique()),
 "cv":"5-fold StratifiedGroupKFold by participant; no participant crosses train CV folds",
 "selection":"participant-aggregated OOF F1, then balanced accuracy, then ROC-AUC",
 "final_validation":"official EATD validation split only after selection",
 "important_limit":"EATD outcome is SDS-derived depression/distress; not direct suicide risk."
}
summary={
 "config":config,
 "training_grouped_oof_participant_metrics":best_oof,
 "official_validation_sample_level_metrics":val_sample_metrics,
 "official_validation_participant_aggregated_metrics":val_part_metrics,
```

6.5 Unified Inference Pipeline

```
import json
from pathlib import Path
import numpy as np
import pandas as pd
import joblib
import librosa

BASE = Path(__file__).resolve().parents[1]
TEXT_MODEL_DIR = BASE / "models" / "text_model"
SPEECH_MODEL_DIR = BASE / "models" / "speech_model"
CONFIG_DIR = BASE / "config"

_TEXT_VEC = None
_TEXT_MODEL = None
_SPEECH_MODEL = None
_SPEECH_CFG = None
_FUSION_CFG = None

def _load_artifacts():
    global _TEXT_VEC, _TEXT_MODEL, _SPEECH_MODEL, _SPEECH_CFG, _FUSION_CFG
    if _TEXT_VEC is None:
        _TEXT_VEC = joblib.load(TEXT_MODEL_DIR / "tfidf_vectorizer.joblib")
    if _TEXT_MODEL is None:
        _TEXT_MODEL = joblib.load(TEXT_MODEL_DIR / "text_svm_calibrated.joblib")
    if _SPEECH_MODEL is None:
        _SPEECH_MODEL = joblib.load(SPEECH_MODEL_DIR / "best_speech_single_audio_model.joblib")
    if _SPEECH_CFG is None:
        _SPEECH_CFG = json.loads((SPEECH_MODEL_DIR / "speech_single_audio_config.json").read_text(encoding="utf-8"))
    if _FUSION_CFG is None:
        _FUSION_CFG = json.loads((CONFIG_DIR / "fusion_config.json").read_text(encoding="utf-8"))

def predict_text(text: str):
    _load_artifacts()
    text = str(text or "").strip()
    if not text:
        raise ValueError("Text input is empty.")
    x = _TEXT_VEC.transform([text])
    p = float(_TEXT_MODEL.predict_proba(x)[0, 1])
    return {
        "text_probability": p,
        "text_binary_prediction": int(p >= 0.5),
        "model": _FUSION_CFG["text_model"],
    }

def extract_audio_features(audio_path):
    audio_path = str(audio_path)
    y, sr = librosa.load(audio_path, sr=16000, mono=True)
    if y.size == 0:
        raise ValueError("Audio contains no samples.")
    mx = float(np.max(np.abs(y)))
    if mx > 0:
        y = y / mx
    mfcc = librosa.feature.mfcc(y=y, sr=sr, n_mfcc=13)
    rms = librosa.feature.rms(y=y)[0]
    zcr = librosa.feature.zero_crossing_rate(y)[0]
    centroid = librosa.feature.spectral_centroid(y=y, sr=sr)[0]
    pitch = librosa.yin(y, fmin=50, fmax=500, sr=sr)
    pitch = pitch[np.isfinite(pitch)]
    f = {
        "duration_sec": float(len(y) / sr),
        "rms_mean": float(np.mean(rms)),
        "rms_std": float(np.std(rms)),
        "zcr_mean": float(np.mean(zcr)),
        "zcr_std": float(np.std(zcr)),
        "spectral_centroid_mean": float(np.mean(centroid)),
        "spectral_centroid_std": float(np.std(centroid)),
        "pitch_mean": float(np.mean(pitch)) if pitch.size else np.nan,
        "pitch_std": float(np.std(pitch)) if pitch.size else np.nan,
    }
    for i in range(13):
        f[f"mfcc{i+1}_mean"] = float(np.mean(mfcc[i]))
        f[f"mfcc{i+1}_std"] = float(np.std(mfcc[i]))
    return f

def predict_speech(audio_path):
    _load_artifacts()
    feats = extract_audio_features(audio_path)
    cols = _SPEECH_CFG["all_input_features"]
    row = pd.DataFrame([[feats.get(c, np.nan) for c in cols]], columns=cols)
    p = float(_SPEECH_MODEL.predict_proba(row)[0, 1])
    thr = float(_SPEECH_CFG["threshold"])
    return {
        "speech_distress_probability": p,
        "speech_binary_prediction": int(p >= thr),
        "speech_threshold": thr,
        "model": _FUSION_CFG["speech_model"],
        "features": feats,
    }

def screening_band(score: float):
    _load_artifacts()
    low = float(_FUSION_CFG["low_upper_exclusive"])
    mod = float(_FUSION_CFG["moderate_upper_exclusive"])
    if score < low:
        return "Low"
    if score < mod:
        return "Moderate"
    return "High"

def combine_scores(text_probability=None, speech_probability=None):
    _load_artifacts()
    if text_probability is None and speech_probability is None:
        raise ValueError("At least one modality score is required.")
```

```
    if text_probability is not None and speech_probability is not None:
        tw = float(_FUSION_CFG["text_weight"])
        sw = float(_FUSION_CFG["speech_weight"])
        score = tw * float(text_probability) + sw * float(speech_probability)
        mode = "text+speech"
        weights = {"text": tw, "speech": sw}
    elif text_probability is not None:
        score = float(text_probability)
        mode = "text-only"
        weights = {"text": 1.0, "speech": 0.0}
    else:
        score = float(speech_probability)
        mode = "speech-only"
        weights = {"text": 0.0, "speech": 1.0}
    return {
        "mode": mode,
        "combined_evidence_score": score,
        "screening_level": screening_band(score),
        "weights": weights,
        "status": _FUSION_CFG["status"],
        "rationale": _FUSION_CFG["rationale"],
    }

def analyze(text=None, audio_path=None):
    t = predict_text(text) if text and str(text).strip() else None
    s = predict_speech(audio_path) if audio_path else None
    f = combine_scores(
        text_probability=t["text_probability"] if t else None,
        speech_probability=s["speech_distress_probability"] if s else None,
    )
    return {
        "text": t,
        "speech": s,
        "decision": f,
        "disclaimer": (
            "Research screening prototype only. The text model was trained on SDCNL suicide/depression Reddit text. "
            "The speech model was trained on EATD SDS-derived depression/distress audio, not direct suicide labels. "
            "The modalities are not participant-matched, so the combined score is a documented prototype late-fusion signal, "
            "not a clinically validated suicide-risk probability or diagnosis."
        ),
    }

if __name__ == "__main__":
    import argparse
    parser = argparse.ArgumentParser(description="Suicide-risk NLP + speech research screening inference")
    parser.add_argument("--text", default=None)
    parser.add_argument("--audio", default=None)
```

6.6 Streamlit User Interface

```
from pathlib import Path
import sys, tempfile
import streamlit as st

BASE = Path(__file__).resolve().parent
sys.path.insert(0, str(BASE / "src"))
from inference_pipeline import analyze

st.set_page_config(
    page_title="Suicide-Risk Detection Using NLP-Based Text and Speech Analysis",
    page_icon=" ",
    layout="centered",
)

st.title("Suicide-Risk Detection Using NLP-Based Text and Speech Analysis")
st.caption("Research screening prototype — NLP text signal + speech distress signal + documented late fusion")

mode = st.radio("Input mode", ["Text only", "Speech only", "Text + Speech"], horizontal=True)

text_value = None
audio_file = None

if mode in ("Text only", "Text + Speech"):
    text_value = st.text_area(
        "Text input",
        height=160,
        placeholder="Enter the text to analyze...",
    )

if mode in ("Speech only", "Text + Speech"):
    audio_file = st.file_uploader(
        "Speech input",
        type=["wav", "mp3", "m4a", "ogg", "flac"],
        help="For closest consistency with EATD training data, WAV speech is preferred.",
    )
    if audio_file is not None:
        st.audio(audio_file)

if st.button("Analyze", type="primary", use_container_width=True):
    if mode == "Text only" and not (text_value or "").strip():
        st.error("Please enter text.")
        st.stop()
    if mode == "Speech only" and audio_file is None:
        st.error("Please upload speech audio.")
        st.stop()
    if mode == "Text + Speech" and (not (text_value or "").strip() or audio_file is None):
        st.error("Please provide both text and speech audio.")
        st.stop()

    tmp_path = None
    try:
        if audio_file is not None:
            suffix = Path(audio_file.name).suffix or ".wav"
            with tempfile.NamedTemporaryFile(delete=False, suffix=suffix) as tmp:
                tmp.write(audio_file.getbuffer())
                tmp_path = tmp.name

        result = analyze(
            text=text_value if mode != "Speech only" else None,
            audio_path=tmp_path if mode != "Text only" else None,
        )

        st.subheader("Screening result")
        d = result["decision"]
        c1, c2 = st.columns(2)
        c1.metric("Research screening level", d["screening_level"])
        c2.metric("Combined evidence score", f"{d['combined_evidence_score']:.3f}")

        if result["text"] is not None:
            st.metric("Text suicide-related score (P_text)", f"{result['text']['text_probability']:.3f}")
        if result["speech"] is not None:
            st.metric("Speech depression/distress score (P_speech)", f"{result['speech']['speech_distress_probability']:.3f}")

        with st.expander("Technical details"):
            st.write("Mode:", d["mode"])
            st.write("Fusion weights:", d["weights"])
            if result["speech"] is not None:
                st.write("Speech binary threshold:", result["speech"]["speech_threshold"])
                st.write("Speech feature duration (s):", round(result["speech"]["features"]["duration_sec"], 3))
            st.write("Decision-layer status:", d["status"])
            st.write("Fusion rationale:", d["rationale"])

        if d["screening_level"] == "High":
            st.warning(
                "This result is not a diagnosis. If a person may be in immediate danger or at risk of self-harm, "
                "contact local emergency services or a qualified mental-health professional."
            )

        st.info(result["disclaimer"])

    except Exception as e:
        st.exception(e)
    finally:
        if tmp_path:
            try:
                Path(tmp_path).unlink(missing_ok=True)
            except Exception:
                pass

st.divider()
st.caption(
    "Current verified text model: TF-IDF + calibrated Linear SVM. "
    "Speech branch: EATD acoustic Logistic Regression selected using participant-grouped cross-validation. "
```

```
    "The combined decision layer is a research prototype and is not clinically calibrated."
)
```

6.7 Automated Verification

```
import json, sys, py_compile
from pathlib import Path
import pandas as pd

BASE=Path(__file__).resolve().parents[1]
sys.path.insert(0,str(BASE/"src"))
results={"checks":[]}
def check(name,cond,detail=""):
    results["checks"].append({"name":name,"pass":bool(cond),"detail":str(detail)})
    if not cond:
        print("FAIL",name,detail)
    else:
        print("PASS",name,detail)

required=[
 BASE/"app.py",
 BASE/"requirements.txt",
 BASE/"README.md",
 BASE/"config"/"fusion_config.json",
 BASE/"src"/"inference_pipeline.py",
 BASE/"models"/"text_model"/"tfidf_vectorizer.joblib",
 BASE/"models"/"text_model"/"text_svm_calibrated.joblib",
 BASE/"models"/"speech_model"/"best_speech_single_audio_model.joblib",
 BASE/"models"/"speech_model"/"speech_single_audio_config.json",
 BASE/"reports"/"text_model"/"text_svm_metrics.json",
 BASE/"reports"/"speech_model"/"speech_single_audio_summary.json",
 BASE/"diagrams"/"Thanmayi_Final_System_Design_Diagrams.drawio",
 BASE/"diagrams"/"DIAGRAM_VERIFICATION_REPORT.json",
]
for p in required:
    check(f"required file: {p.relative_to(BASE)}",p.exists(),p)

# Python syntax
for p in list((BASE/"src").glob("*.py"))+[BASE/"app.py"]:
    try:
        py_compile.compile(str(p),doraise=True)
        check(f"python syntax: {p.name}",True)
    except Exception as e:
        check(f"python syntax: {p.name}",False,e)

# Dataset integrity
tr=pd.read_csv(BASE/"Datasets"/"training-set-clean.csv")
te=pd.read_csv(BASE/"Datasets"/"testing-set-clean-leakage-free.csv")
check("text train rows",len(tr)==1513,len(tr))
check("text test rows",len(te)==379,len(te))
check("train labels binary",set(tr["is_suicide"].astype(int).unique())<= {0,1},sorted(tr["is_suicide"].unique()))
check("test labels binary",set(te["is_suicide"].astype(int).unique())<= {0,1},sorted(te["is_suicide"].unique()))
norm=lambda s:" ".join(str(s).casefold().split())
overlap=set(map(norm,tr["model_text"])) & set(map(norm,te["model_text"]))
check("train-test normalized text overlap = 0",len(overlap)==0,len(overlap))

# Model metrics integrity
tm=json.loads((BASE/"reports"/"text_model"/"text_svm_metrics.json").read_text())
check("text metric accuracy exact",abs(tm["accuracy"]-0.7255936675461742)<1e-12,tm["accuracy"])
sm=json.loads((BASE/"reports"/"speech_model"/"speech_single_audio_summary.json").read_text())
vm=sm["official_validation_participant_aggregated_metrics"]
check("speech validation participant count",sm["config"]["validation_participants"]==79,sm["config"]["validation_participants"])
check("speech validation metrics present",all(k in vm for k in ["accuracy","balanced_accuracy","precision","recall","f1_score","roc_auc"]),vm)

# Diagrams
dr=json.loads((BASE/"diagrams"/"DIAGRAM_VERIFICATION_REPORT.json").read_text(encoding="utf-8"))
check("six diagrams present",dr["page_count"]==6,dr["page_count"])
check("diagram structural verification pass",dr["pass"] is True,dr["pass"])

# Inference smoke tests
from inference_pipeline import analyze, predict_text, predict_speech, combine_scores
neutral=predict_text("I had a normal day and spent time with my friends.")
check("text inference probability range",0<=neutral["text_probability"]<=1,neutral["text_probability"])

# pick a valid EATD validation wav
audio=None
for p in sorted((BASE/"Datasets"/"validation").glob("*/*_out.wav")):
    if p.stat().st_size>100:
        audio=p; break
check("validation audio found",audio is not None,audio)
if audio:
    sp=predict_speech(audio)
    check("speech inference probability range",0<=sp["speech_distress_probability"]<=1,sp["speech_distress_probability"])
    fused=combine_scores(neutral["text_probability"],sp["speech_distress_probability"])
    check("fusion score range",0<=fused["combined_evidence_score"]<=1,fused)
    check("fusion screening level",fused["screening_level"] in {"Low","Moderate","High"},fused["screening_level"])

results["passed"]=sum(c["pass"] for c in results["checks"])
results["failed"]=sum(not c["pass"] for c in results["checks"])
results["overall_pass"]=results["failed"]==0
out=BASE/"reports"/"FINAL_AUTOMATED_TEST_REPORT.json"
out.write_text(json.dumps(results,indent=2,ensure_ascii=False),encoding="utf-8")
print(json.dumps({"passed":results["passed"],"failed":results["failed"],"overall_pass":results["overall_pass"]},indent=2))
sys.exit(0 if results["overall_pass"] else 1)
```

CHAPTER - 7

7. TESTING

The automated verification suite checks files, syntax, datasets, metrics, diagrams and branch inference.

7.1 Types of Testing Implemented

Unit Testing: individual functions.

Integration Testing: model and UI interaction.

Dataset Integrity Testing: row counts, labels and leakage.

System Testing: complete app.

Diagram Verification: six required diagrams.

7.2 Test Cases and Results

ID	Test Description	Expected Output	Result
TC1	Required project files	Files/models/configs available	Pass
TC2	Python source syntax	All source files compile	Pass
TC3	Text train rows	1,513 rows	Pass
TC4	Text test rows	379 rows	Pass
TC5	Train-test overlap	0 normalized overlaps	Pass
TC6	Text inference	Probability in [0,1]	Pass
TC7	Speech inference	Real EATD WAV returns probability	Pass
TC8	Fusion output	Score in [0,1] + valid band	Pass
TC9	Diagram verification	6 pages, no structural errors	Pass
TC10	Complete automated suite	37 passed, 0 failed	Pass

CHAPTER - 8

8. RESULTS AND EVALUATION

The calibrated TF-IDF SVM achieved the strongest verified text results on the common held-out test set.

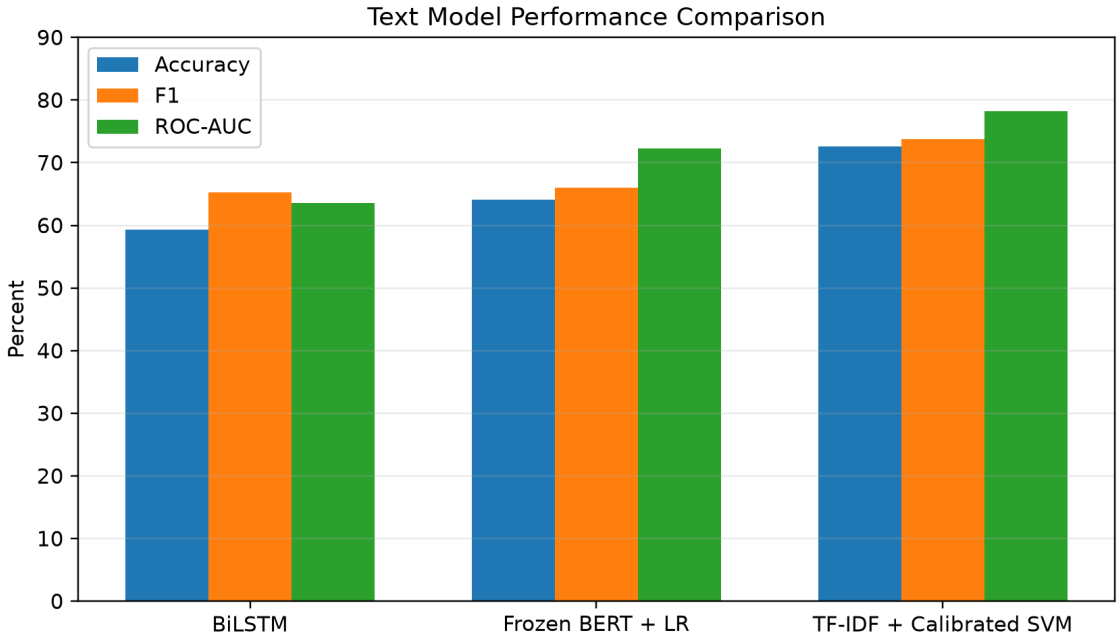
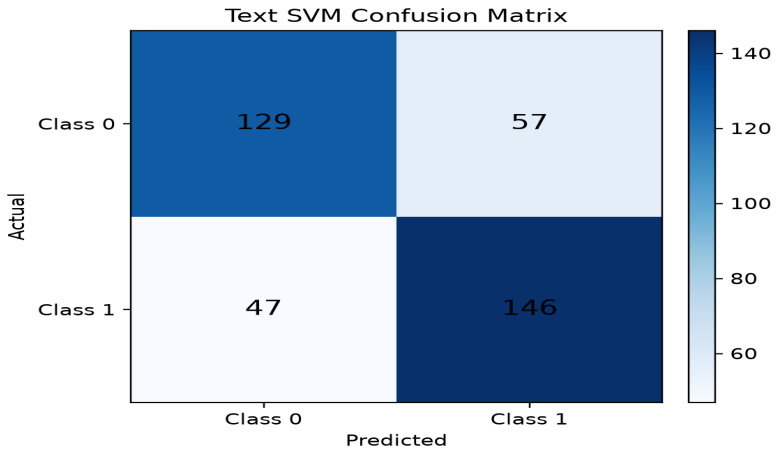


Figure 8.1: Text Model Performance Comparison

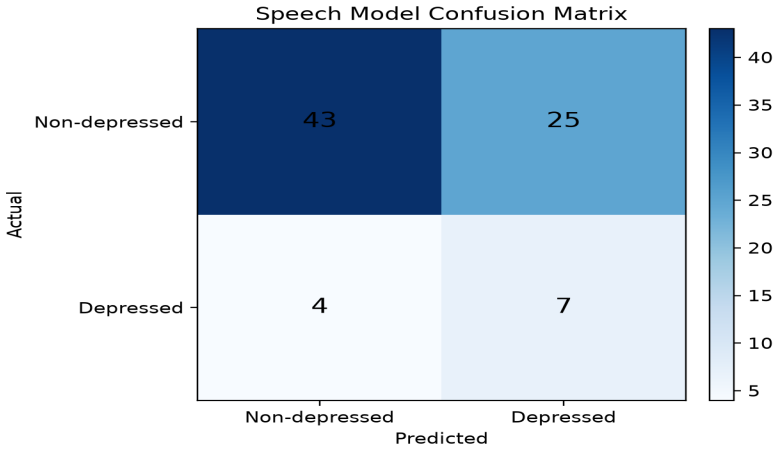
8.1 Text Results

Model	Accuracy	Precision	Recall	F1	ROC-AUC
BiLSTM	59.37	57.77	75.13	65.32	63.58
Frozen BERT + LR	64.12	63.77	68.39	66.00	72.27
TF-IDF + SVM	72.56	71.92	75.65	73.74	78.22

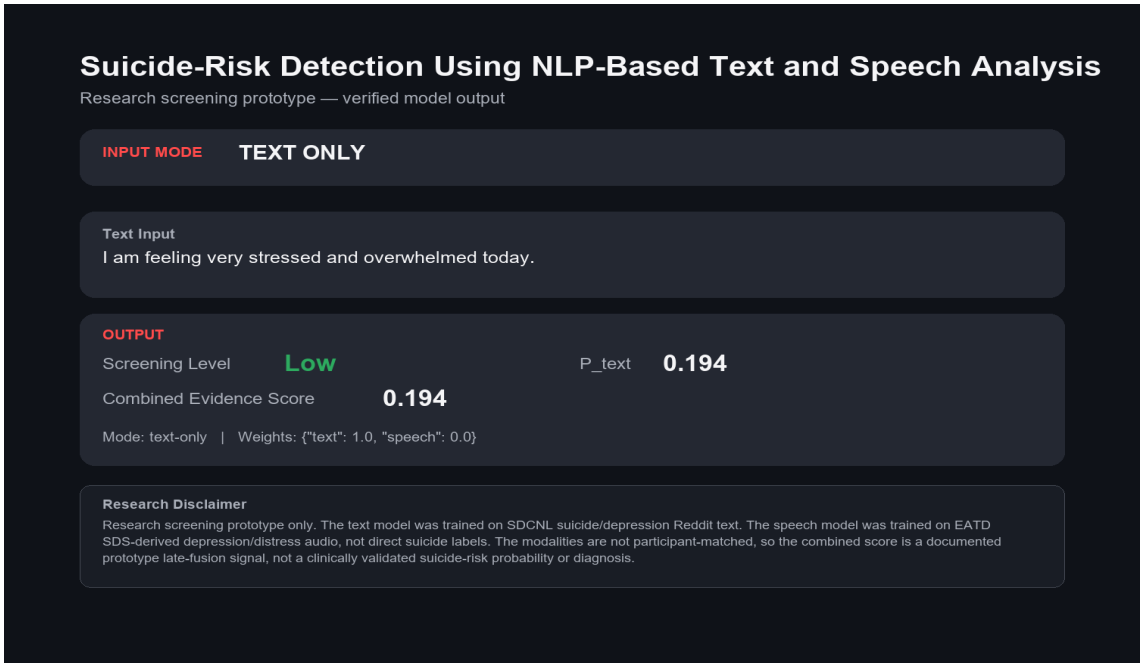


8.2 Speech Results

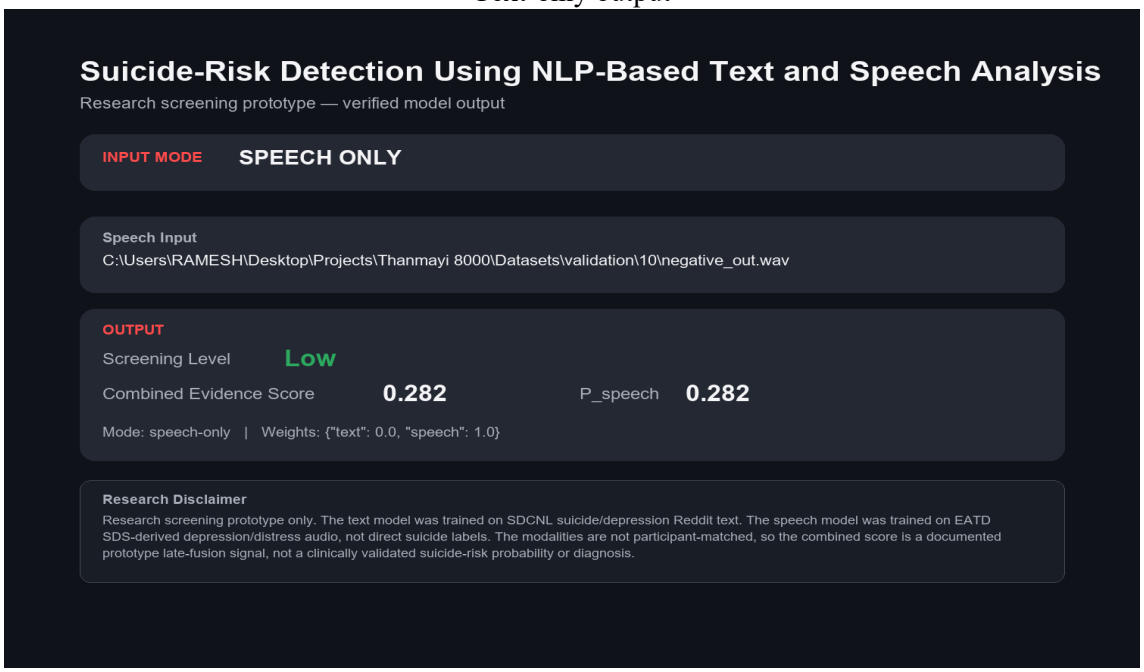
Participant-level EATD validation: accuracy 63.29%, balanced accuracy 63.44%, precision 21.88%, recall 63.64%, F1 32.56%, ROC-AUC 59.22%.



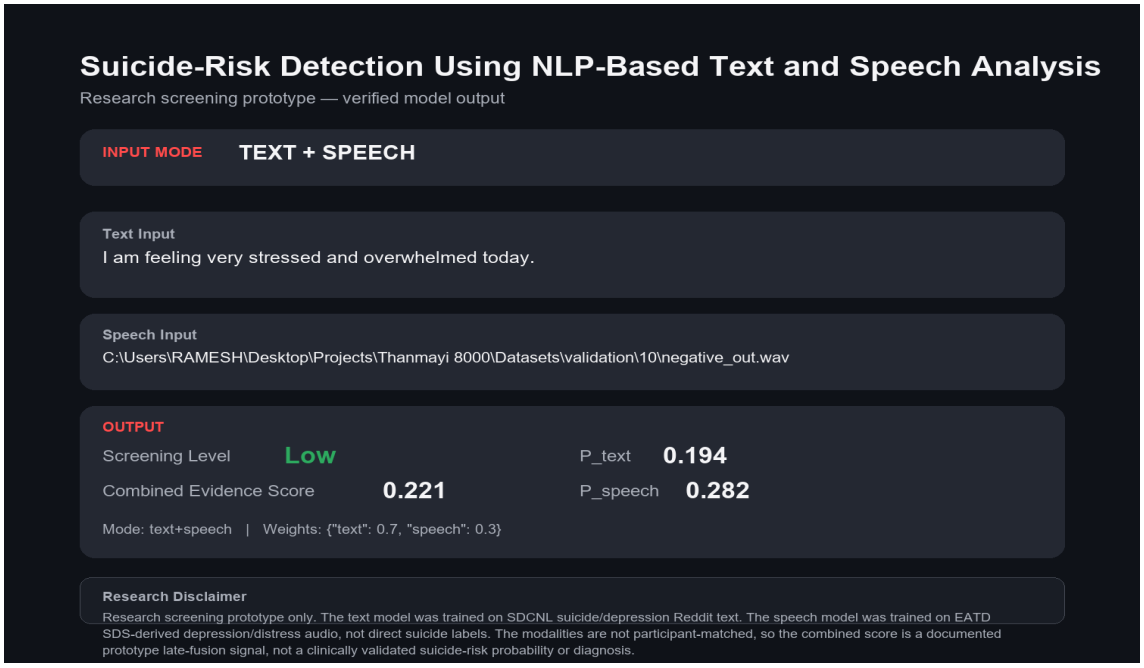
8.3 Application Output



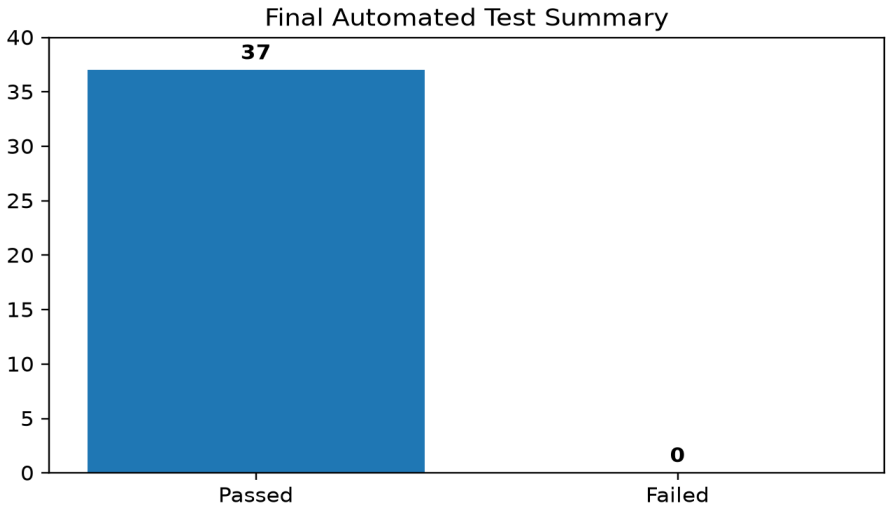
Text-only output



Speech-only output



Text + Speech output



Final automated verification: 37 passed, 0 failed

CHAPTER - 9

9. CONCLUSION

The project implemented a complete modular research-screening prototype using NLP text analysis and complementary acoustic speech analysis.

TF-IDF + calibrated Linear SVM is the best verified text model. The speech model is reported with modest validation performance and is not overstated.

The application supports all three requested input modes and passed the final automated test suite.

Because the datasets are unmatched, the combined score is a transparent prototype evidence score and not a clinically validated suicide-risk probability.

CHAPTER - 10

10. FUTURE ENHANCEMENT

Collect or access paired multimodal suicide-labelled data.

Improve multilingual and suicide-specific speech modelling.

Evaluate fully fine-tuned transformers under leakage-free testing.

Add secure, consent-aware data management and independent clinical validation before any real healthcare deployment.

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APPENDIX

Suicide-Risk Detection Using NLP-Based Text and Speech Analysis

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Abstract— This article presents the verified current implementation of the project, including leakage-free SDCNL text modelling, EATD acoustic speech modelling and transparent prototype late fusion.

I. INTRODUCTION The aim is to support academic research into multimodal screening without overclaiming clinical validity.

II. METHODOLOGY Independent text and speech branches are evaluated separately and combined only at the final decision layer.

III. RESULTS The best text model achieved 72.56% accuracy and 78.22% ROC-AUC; the final automated verification suite passed 37/37 checks.

IV. CONCLUSION The implementation is complete as a research prototype; clinical multimodal validation remains outside the evidence supported by the unmatched datasets.

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